

OXFORD

INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL A-LEVEL BIOLOGY

(9610) BL03

Report on the examination

January 2024

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL BIOLOGY (9610) BL03 JANUARY 2024

Good answers were seen to all questions with some students demonstrating excellent knowledge and understanding of the specification content. Students had obviously referred to previous exam papers and mark schemes to help them prepare for the exam.

Plotting a graph caused problems with a very small percentage of students gaining all available marks.

Information provided should be read carefully and used to answer the questions, but marks are not awarded for repeating information already provided.

Students should link their answers to the context of the question and not rely on rote learned phrases such as 'small sample size' or 'correlation does not mean causation'.

QUESTION 01

This question assessed the principles of nutrient cycling in the context of the phosphorous cycle. Some students ignored this context and wrote about the nitrogen cycle, or just repeated information provided in Figure 1.

01.1

To gain marking point 1, reference to both feeding and digestion were needed; better answers used 'hydrolyse' instead of digest. Although the term 'assimilation' is used in the specification, it was unusual for it to be included in student responses. Many students did demonstrate an understanding of assimilation in their descriptions of the incorporation of absorbed molecules into named organic molecules in consumers.

A number of answers described the use of phosphorus to produce amino acids; students should be reminded that, although proteins can be phosphorylated, amino acids do not contain phosphorus.

01.2

Students were provided with guidance in the form of information in Figure 1, but just repeating this information in the answer was insufficient to gain any marking points. For marking point 2, phosphorus release was not accepted, and answers had to specify that ions or simple inorganic molecules were produced. For this reason, this marking point was often missed by students. Better answers clearly and concisely linked the three marking points, with a small percentage of students gaining full marks. The best answers seen used good A-level terminology throughout, including saprophytic, hydrolyse, and absorb.

QUESTION 02

This question was about aerobic and anaerobic respiration and included analysis of data.

02.1

This was a well-answered questions with the majority of students gaining 1 mark for correctly stating where glycolysis occurs. Occasional incorrect answers included mitochondria, matrix, granum, and stroma.

02.2

Another well-answered question with most students identifying substance X as phosphorylated glucose. Incorrect answers included glycerate phosphate and pyruvate.

02.3

The majority of students correctly added the arrow showing where ATP is produced. In a number of scripts, this question was not attempted and students should be reminded to read the paper carefully as not all questions have an answer line.

02.4

Most students showed an understanding of the concept of 'net gain', with better answers stating that 2 ATP are used in glycolysis but 4 are produced, giving an overall gain of 2 ATP. Some answers just said, 'what is left after a reaction' and were too vague to be awarded the mark.

02.5

Many incorrect answers of either chi-squared or correlation coefficient were given. Most of the correct answers named the t-test rather than standard error and 95% confidence limits. Students are expected to know when to use the statistical tests listed in the mathematical requirements section of the specification.

02.6

Many students identified the overlap in some of the standard error bars, but they also had to realise what this meant; saying 'the results are not significant' did not gain the mark. The other most common correct answers included 'only one concentration of lactate', and 'only muscle from rat'.

Some generic answers like 'correlation does not mean causation' did not gain marks. Other answers did not focus on the validity of the results, for example, stating that the small sample size means the results are not representative.

02.7

The instruction given in the question to 'Use data from Figure 4' meant that students could not gain full marks without correctly reading values from the graph. Data should be used to support the trends given, for example, 'as running distance increases, the aerobic energy supply increases 24 times from 10 to 240 arbitrary units'. Although students did not lose a mark for 'energy produced' they should know that this is incorrect.

02.8

Some students who did not gain marks for this ratio calculation used the wrong data from the graph (reading off energy supply from 1500m rather than 800m). Some gave their answers as fractions rather than decimals. Some seemed to struggle to give their answers in the order 'aerobic:anaerobic' with anaerobic being 1.

02.9

This question proved surprisingly challenging, with many answers lacking the required detail or repeating the stem of the question. The guidance in the question was that athletes run very quickly at 200m, so the marking points required the idea of rate: high rate of muscle contraction, and oxygen not supplied to muscles fast enough.

QUESTION 03

This question about the efficiency of food production involved the evaluation of data. Most students were able to access this question and gain some marks.

03.1

A few students incorrectly read values from the graph eg 12.5% for chicken and so calculated the wrong value. A few incorrectly added a % symbol to their answer. Some students just found the difference between the two percentages.

03.2

This question was well answered with students showing good knowledge of increasing energy efficiency in food production.

03.3

Students were good at analysing and evaluating the information provided, with over 60% scoring 1 or 2 marks; a very low percentage of answers were awarded 0 marks. Some generic statements were seen and did not gain marks including 'correlation does not mean causation' and 'no statistical test'.

All marking points were seen, with most answers including marking point 2. Some answers just stated that the conclusion was incorrect and gave reasons why. An 'evaluate' question requires a balanced answer giving evidence for and against.

QUESTION 04

This question about inheritance was based on hair colour in alpacas.

04.1

Although some good definitions were given for 'locus', many found this question challenging and just over half of students were awarded the mark. Vague answers that did not gain the mark often just referred to the position of a gene without also naming the chromosome.

04.2

Very good understanding of the context of the question was seen and most students gave the two correct genotypes. Where the mark was not awarded, students had not given the alleles for both genes, or had mixed up the two genes.

04.3

A very high percentage of students gave the four correct gametes that could be produced from the genotype AaEe. Incorrect answers tended to split up these four letters to give A, a, E, and e, or split up the two genes to give Aa and Ee.

04.4

Some good, clear answers making links back to the question were seen. There was a high percentage of unattempted questions, and many answers that just repeated the stem of the question. The question did ask students to refer to alleles G and g, so marking points 2 and 3 required references to genotypes as well as phenotypes.

04.5

Lots of good Hardy-Weinberg calculations were seen, with over two thirds of students gaining all three marks. The most common error was identifying 0.295 as q^2 rather than just q . Some rounding errors were also seen eg rounding down to 41%

QUESTION 05

This question was based on required practical 9 investigating the effect of competition on seedling growth. It assessed knowledge of practical design, recording data, graph plotting, and evaluation skills.

05.1

Most students gained the one mark for this question by correctly stating 'intraspecific'. Occasional wrong answers gave interspecific.

05.2

Many of the answers that did not score both marks referred to control variables that the student would not easily be able to control outside, including temperature and light intensity. Although species of carrot and age of seeds were accepted, the size of each seed was not.

05.3

Answers to this question showed that students have a very good understanding of what dry mass is and why it is important to use this value rather than fresh mass. Better answers made a link to the dry mass reflecting the mass of organic material in the plant. Some of the answers that did not gain the mark just stated that fresh mass includes water, or that dry mass does not include water, without giving the reason why it was used.

05.4

It was good to see that the majority of students correctly rounded the values to an appropriate number of decimal places.

05.5

Most students attempted to draw the graph and gained at least 2 marks. A very small percentage gained all 4 marks, with the final marking point commonly lost for not drawing a suitable line of best fit.

It was relatively common to see students choose a suitable y-axis scale then incorrectly plot the first two values at 0.19 rather than 0.18.

05.6

Most students gained marking point 1 for correctly recognising how the data supported the conclusions. Evidence against the conclusion was most commonly for identifying that dry mass was the same at 5 plants, or no intermediates between 10 and 20. The time scale of 12 weeks was also identified as being insufficient time for plant growth.

Some answers that did not gain marks lacked sufficient depth and detail, including 'no repeats' or 'no statistical test'; students should clarify these points. Good answers stated that a statistical test would be used to see if the difference in dry mass was significant, not to see if the 'results were due to chance'. There was some use of generic statements like 'correlation does not mean causation'.

QUESTION 06

This data analysis question was about the effect of carbon dioxide concentration on the rate of photosynthesis and included some recall of the light-dependent reaction.

06.1

This question required the interpretation of a graph showing carbon dioxide concentration against rate of photosynthesis. Many good answers were seen which used the term 'rate' and recognised the balance between photosynthesis and respiration. Some answers showed the common misconception that photosynthesis and respiration do not occur at the same time, or that respiration only occurs at night or when it is dark.

06.2

A lot of really excellent answers were seen that included clear, concise, and correctly used key terms. The most common marking point lost was 3, as answers did not state that GP was reduced to triose phosphate. Some answers did not gain full marks as they described the light-independent reaction without linking it to the context of the question which was the effect of increasing carbon dioxide concentration. It is worth reminding students that they can use just the initials of molecules when this is how they are referred to in the specification including RuBP and GP but should fully write out the names of others including triose phosphate.

06.3

Students seemed to struggle with this calculation of percentage change. Just over half of students gained both marks and many only scored 1 mark for the correct reading of two values from the graph.

06.4

This question identified some misconceptions about limiting factors with answers suggesting that carbon dioxide was the limiting factor when it was at high concentrations. Some answers did not gain any marks as they just stated the limiting factors at each concentration without linking these to the differences in percentage changes in rate.

QUESTION 07

This final question on the paper required extended responses and there were some really good answers showing excellent understanding and using correct terminology throughout.

07.1

Common marking points recognised that electrons pass down the electron transfer chain and the energy released is used to pump protons into the intermembrane space; the role of ATPase was also well-known. The direction of proton movement and the role of mitochondrial membranes was less well understood; some answers referred to protons being pumped **into** the inner mitochondrial membrane, and some seemed to confuse the relative positions of the matrix and the intermembrane space.

Some answers wasted time and space by writing about glycolysis when the question only asked for ATP synthesis **inside** the mitochondria.

Some superb answers were seen and about one fifth of students gained the maximum 5 marks.

07.2

This question gave a clear structure to guide students and better answers made good use of the structure by bullet pointing their answers to match the question. There were lots of good answers in which students made clear links between their knowledge of aerobic respiration and the structure of mitochondria, but a very small percentage gained the maximum 6 marks.

07.3

Some students misinterpreted this question and wrote about speciation or even succession. This question was unattempted by about 10% of students.

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